

Supplemental information

Determination of gene-level fitness contributions across the mucin-utilization network of *Bacteroides thetaiotaomicron*

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Supplemental Text

Text S1. The three-tiered mucin utilization network in *B. thetaiotaomicron*

Initial engagement of host-derived mucin glycans is likely shaped by the cell-surface architecture of *B. thetaiotaomicron*. Surface glycan-binding proteins, including SusD-like proteins, TonB-dependent transporters and CBM-containing enzymes, can promote substrate sensing, retention and access of mucin-derived glycan motifs at the outer membrane [1-3]. In parallel, capsular polysaccharides may indirectly influence mucin access by altering the steric properties of the cell surface [1] (**Figure. 1a**). Although direct evidence for CPS-mediated control of mucin access remains limited, the coordinated regulation of host-glycan-responsive PULs and capsule biosynthesis suggests that capsule remodeling is integrated with glycan-acquisition programs [4]. Following substrate encounter, mucin processing proceeds through coordinated activation of PULs, which couple glycan sensing and TonB-dependent transport with the activities of CAZymes (**Figure. 1b**) In particular, glycosidic hydrolases, sulfatases and mucin-associated peptidases that degrade mucin *O*-glycans into smaller oligo-, mono-saccharides and glycopeptides [5-8]. The resulting complex oligosaccharides are then captured and imported into the periplasm via TonB-dependent transporters, where a consortium of CAZymes further depolymerize them into transportable mono- and disaccharides. *Bacteroides* utilizes mucin-derived sugars liberated from mucin, where major constituents include fucose, galactose, *N*-acetylglucosamine (GlcNAc) and *N*-acetylgalactosamine (GalNAc) [9]. These liberated monosaccharides are routed through highly specific, dedicated catabolic pathways [3, 10], and *B. thetaiotaomicron* is predicted to encode each of this catabolic pathway (**Figure. 1c**). For instance, it encodes a fucose utilization operon that enable isomerization and phosphorylation of fucose before entering central metabolism [11, 12]. In addition, it also encodes the Leloir pathway, as well as genes associated with transport of hexosamines into glycolysis [13]. As the result, it is expected that *B. thetaiotaomicron* utilize these catabolic pathways funnel carbon into the central metabolic pathways.

Text S1. Condition specific fitness profile in genes associated with amino acid biosynthetic pathways.

Gene-level analysis further revealed condition-specific fitness dependencies In MM-glc, genes associated with methionine biosynthesis (BT_0180) and those that channel the glycolytic intermediate 3-phosphoglycerate into *de novo* serine biosynthesis (BT_1152 and BT_1153) were more extensively depleted (**Table S5**). Conversely, in MM-muc, genes involved in L-homoserine biosynthesis and degradation (BT_2403, BT_2413, and BT_3636) showed stronger depletion in MM-muc. Together, these results highlight condition-specific amino acid dependencies in *B. thetaiotaomicron*, which recapitulates the previous observation that the specific metabolic bottlenecks and pathway dependencies undergo significant shifts depending on the primary carbon source [14].

Text S2. MM-muc specific fitness defect in BT_0771

Despite the broad functional complementation observed across most GHs, BT_0771 which encodes a GH13 glucan-branching enzyme, showed a notable fitness defect (z-score < -1.0) (**Figure. 5a**). Genomic context analysis revealed that BT_0771 resides within the BT_0769-0771 operon, positioned immediately downstream of the well-characterized small RNA, *gibS* (**Figure. S9a**). Consistent with our expression data, *gibS* is known to be induced by GlcNAc, mucin, or entry into the stationary phase, functioning as a repressor for specific downstream targets, including BT_0771 [15, 16]. Intriguingly, our CRISPRi results and previous transposon-insertion sequencing data [14] demonstrate that targeted knockdown or inactivation of BT_0771 leads to a fitness decline not only in MM-muc, but across a wide range of carbon-supplemented defined media (**Figure. S9b, c**). Together, these results point toward an intricate regulatory network. While the expression of BT_0771 is tightly repressed by *gibS* in the presence of glycans and their constituent sugars, maintaining a basal level of this enzyme appears to retain a non-trivial contribution to fitness across multiple physiological conditions.

Text S3. iModulon-based view of condition-dependent transcriptional remodeling between MM-glc and MM-muc.

To confirm whether such condition-specific gene essentiality translates to transcriptome-level changes, we referred to *B. thetaiotaomicron* iModulon framework [17]. iModulons are statistically inferred gene modules whose members covary across transcriptomes and therefore often reflect shared regulation or common physiological function [18, 19]. By projecting gene-level expression changes into module activities, iModulon analysis provides a systems-level view of how cellular programs are reallocated between conditions.

Differential iModulon activity analysis comparing MM-glc and MM-muc revealed a particularly strong remodeling among PUL-associated modules. Eight PUL iModulons was above the iModulon threshold of 3 in MM-muc, whereas only one PUL iModulon fell below the negative threshold (**Figure. S9**). Mapping these iModulons onto the pathway, activated PUL modules localized largely to the known polysaccharide-utilization genes (**Figure. S11**). Notably, all activated PUL iModulons except HTCS_Hya, HTCS_Man1_3, and SigM_PUL13_21 belonged to PULs induced by mucin *O*-glycans [5], indicating that the iModulon framework recapitulated the known mucin-responsive transcriptional program of *B. thetaiotaomicron* grown on mucin.

Amino acid metabolism represented a second major axis of transcriptional remodeling between MM-glc and MM-muc (**Figure. S10**). In particular, the Glu/Asp iModulon showed higher activity in MM-muc, whereas the Glu/Gln/Asn, Trp/Tyr, and Val/Leu/Ile iModulons showed lower activity in MM-muc than in MM-glc. Mapping these iModulons onto the pathway context by and large recapitulated the amino acid metabolic pathway (**Figure. S11**). At gene-level, those assigned to the Glu/Asp iModulon showed a significant, uniform up-regulation in MM-muc, whereas genes in the Glu/Gln/Asn iModulon the Trp/Tyr and Val/Leu/Ile iModulons (pathways) were uniformly downregulated (**Figure. S11, Table S6**). Together, these results indicate that the shift from glucose to mucin is accompanied by broad up-regulation of PULs, alongside with a selective redistribution of transcriptional changes across amino acid biosynthetic programs as consistent with the result in **Figure. 3b**. Together, these findings recapitulate previous observations that specific metabolic bottlenecks and

pathway dependencies undergo significant shifts depending on the primary carbon source [14].

Text S4. Experimental setting and quality control for *in vivo* CRISPRi screen.

Mice were treated with antibiotics for clearance of native gut microflora prior to library inoculation. CRISPRi library was allowed to propagate for 3 days before being sacrificed for NGS run (**Figure. 7A**). Read depths of colon samples were comparable to those of *in vitro* (**Table S2**). PCA analysis showed a high degree of similarity between biological replicates (n=7), with the exception of replicate #3 and #4, which separated from the main cluster, and exhibited a lower correlation between replicates (**Figure. S13a, b**). Consistent with this, sgRNA insertion frequency in replicates #3 and #4 displayed perturbed distribution patterns suggestive of over-depletion or technical errors and were omitted from downstream analysis (**Figure. S13c**). The z-score distribution showed a clear depletion on essential gene sets than the rest (**Figure. S13d**) [14], where strong negative fitness effects were concentrated in pathways associated with core growth functions, including translation, ribosome biogenesis, transcription, and cell envelope biogenesis (**Figure. S13e**), supporting the technical and biological fidelity of the colon CRISPRi screen.

Text S5. Quadrant-level analysis of fitness changes between MM-muc and *in vivo*.

We examined functional enrichments across the four quadrants of the fitness scatter plot (**Figure. 7c; Table S8**). Genes depleted in both MM-muc and the colon (Q3) were enriched for translation- and ribosome-associated functions, consistent with shared core growth requirements across the two environments. In contrast, genes displaying discordant fitness profiles between the two environments in Q2 were enriched for glycosyltransferase activity [GO:0016757], which include BT_3366, BT_3367, and BT_3368. Notably, their *in vivo* z-score distribution mirrored that of the MM-glc condition rather than the MM-muc (**Figure. 7c, Table S4**). This suggests that the observed discordance is driven by a mucin-specific fitness requirement rather than a general determinant of colon fitness. Although Q1 and Q4 lacked functional category enrichments, locus-specific analysis revealed distinct phenotypic patterns (**Table S8**). In Q1, highly dispensable gene sets tend to be PULs (or corresponding CAZys) and CPSs, in colon and MM-muc, respectively, highlighting their functional divergence between the two conditions. Q4 on the other hand, comprised a functionally heterogeneous set of genes with divergent fitness effects between the two conditions, implications of which remain unclear.

Supplemental Figures

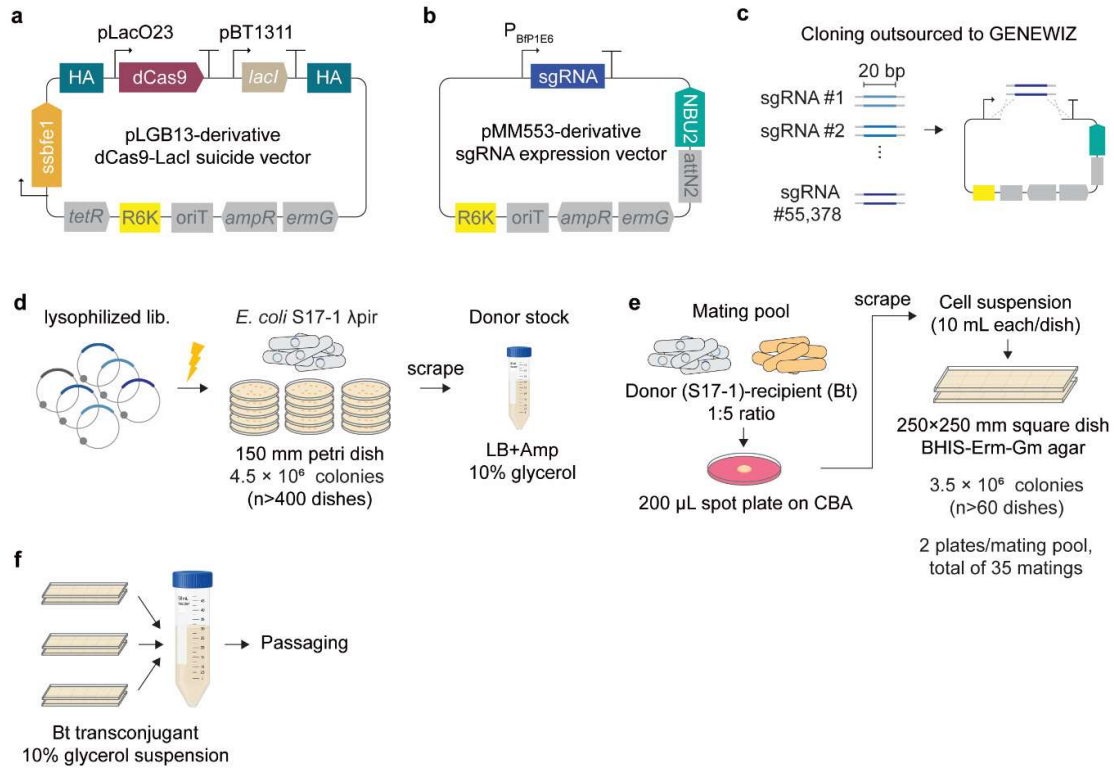


Figure S1. Schematic of the genome-scale CRISPRi library construction in *B. thetaiotaomicron*. (a) Schematic of the pLGB13-derived dCas9–LacI suicide vector used to introduce the inducible CRISPRi machinery into the *B. thetaiotaomicron* chromosome. (b) Schematic of the NBU2-containing, pMM553-derived sgRNA expression vector, in which sgRNA expression is driven by the constitutive P_{BfP1E6} promoter. (c) Construction of the pooled sgRNA plasmid library. A total of 55,378 sgRNAs containing 20-bp gene-targeting sequences were cloned into the sgRNA expression vector. Vector library cloning was performed by GENEWIZ. (d) Generation of the *E. coli* S17-1 λ pir donor library. The lyophilized plasmid library was transformed into S17-1 λ pir and plated on 150-mm LB–ampicillin agar plates, yielding approximately 4.5×10^6 colonies across >400 plates. Colonies were pooled by scraping and preserved in LB containing ampicillin and 10% glycerol to generate the donor stock. (e) Transfer of the pooled sgRNA library into *B. thetaiotaomicron* by conjugation. Donor S17-1 λ pir and recipient *B. thetaiotaomicron* cells were combined at a 1:5 donor-to-recipient ratio, and 200 μ l of each mating mixture was spotted onto CBA plates. Following mating, cells were recovered by scraping, resuspended in 20 mL of BHIS, and plated (10 mL) onto 250 $\times$ 250-mm square dish containing BHIS agar with erythromycin and gentamicin for transconjugant selection. Two selection plates were used per mating pool across 35 independent matings, yielding approximately 3.5×10^6 transconjugant colonies from >60 plates. (f) Selected *B. thetaiotaomicron* transconjugants were pooled, preserved as 10% glycerol suspensions, and used for subsequent passaging and CRISPRi fitness screening.

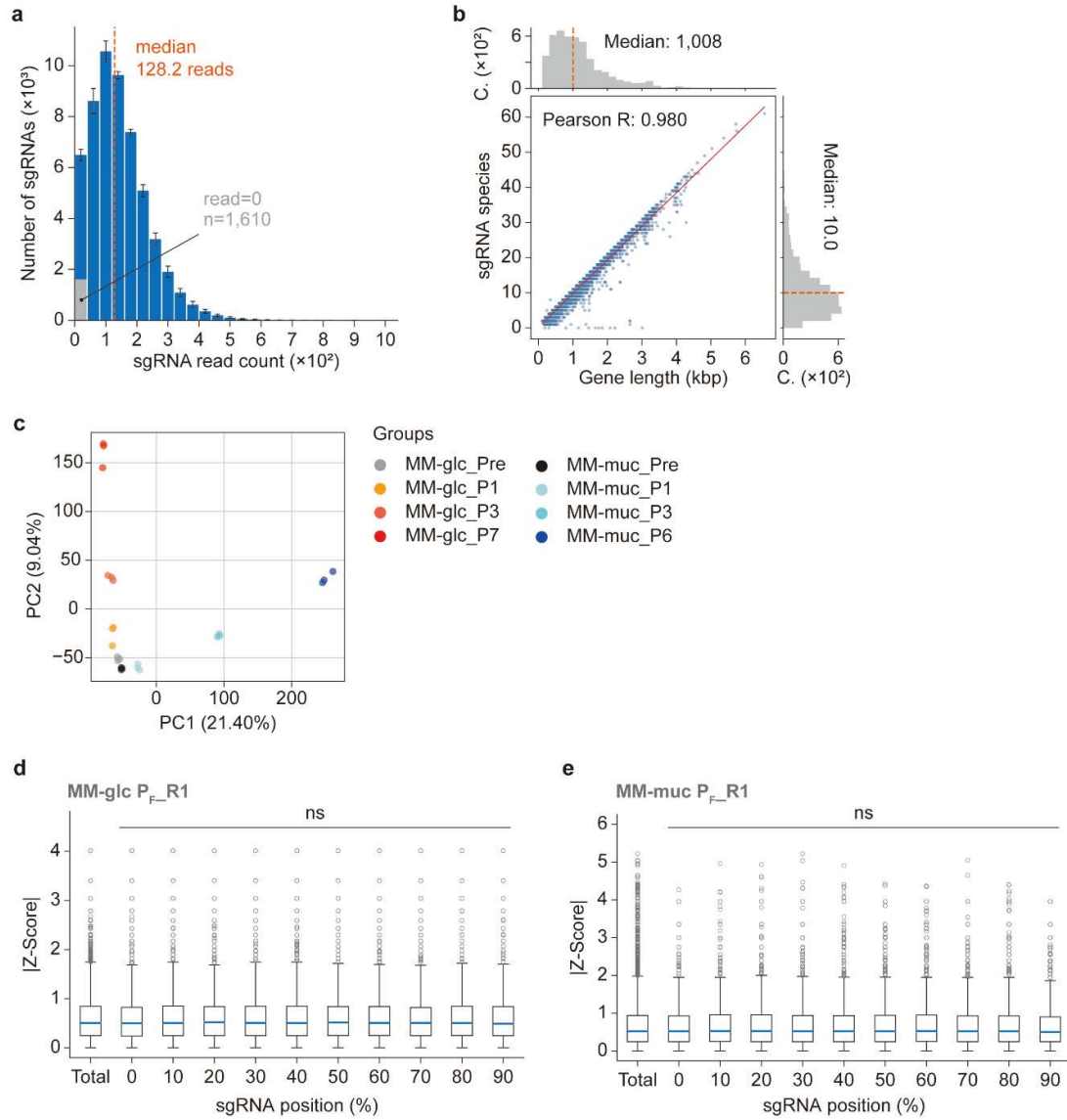


Figure S2. Quality assessment of the genome-scale CRISPRi sgRNA library and positional distribution of sgRNA-derived fitness scores. (a) Distribution of sequencing read counts for individual sgRNAs in the initial CRISPRi library. The orange dashed line indicates the median read count (128.2 reads per sgRNA). sgRNAs with zero reads are indicated in gray. Error bars represent variation across replicate measurements. (b) Correlation between gene length and the number of uniquely mapped sgRNAs per coding sequence. The central scatter plot shows a strong linear correlation between gene length and sgRNA count. Histograms show the distributions of CDS length and sgRNA count per gene; orange dashed lines indicate the respective medians. (c) Principal component analysis (PCA) of genome-wide sgRNA abundance profiles across the initial library and sequential passages in minimal medium containing glucose (MM-glc) or mucin (MM-muc). (d, e) Distribution of

sgRNA-derived z-scores as a function of target position along coding sequences in the final passage of MM-glc **(d)** and MM-muc **(e)** for representative replicate Box plots show the median and interquartile range, with whiskers extending to $1.5\times$ the interquartile range. ns: non-significant.

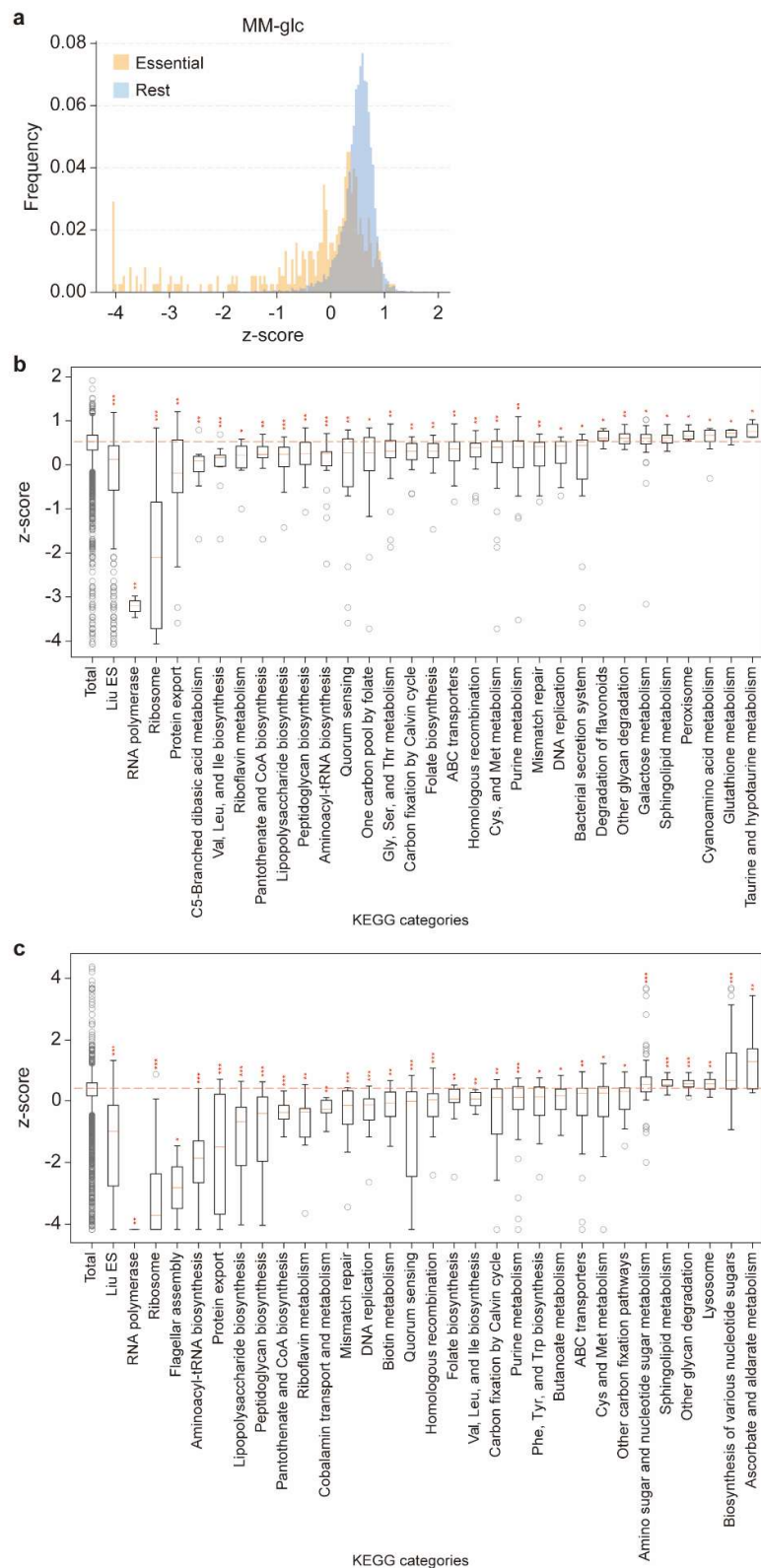


Figure S3. Benchmarking CRISPRi-derived fitness scores against previously defined essential genes and functional categories. (a) Distribution of gene-level fitness z-scores for genes previously classified as essential versus the remainder of the genome in MM-glc. (b,c) Box plots showing the distribution of gene-level z-scores across selected KEGG functional categories in (b) MM-glc and (c) MM-muc. The category labeled Liu ES denotes the set of essential genes defined previously [14].

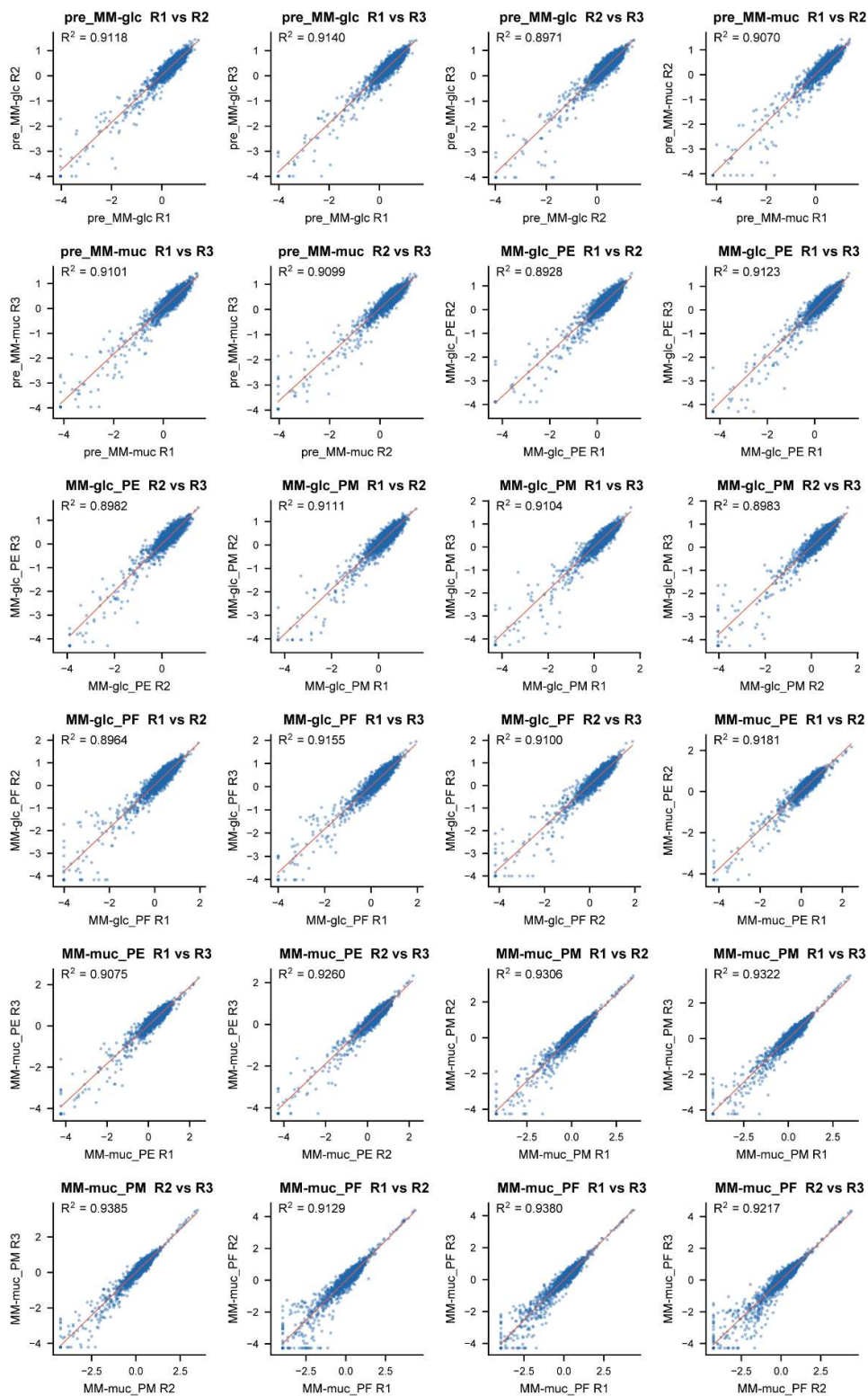


Figure S4. Reproducibility of genome-wide CRISPRi fitness profiles across biological replicates. Pairwise scatter plots comparing gene-level fitness z-scores between biological replicates for the initial library (pre) and for sequential passages in MM-glc or MM-muc.

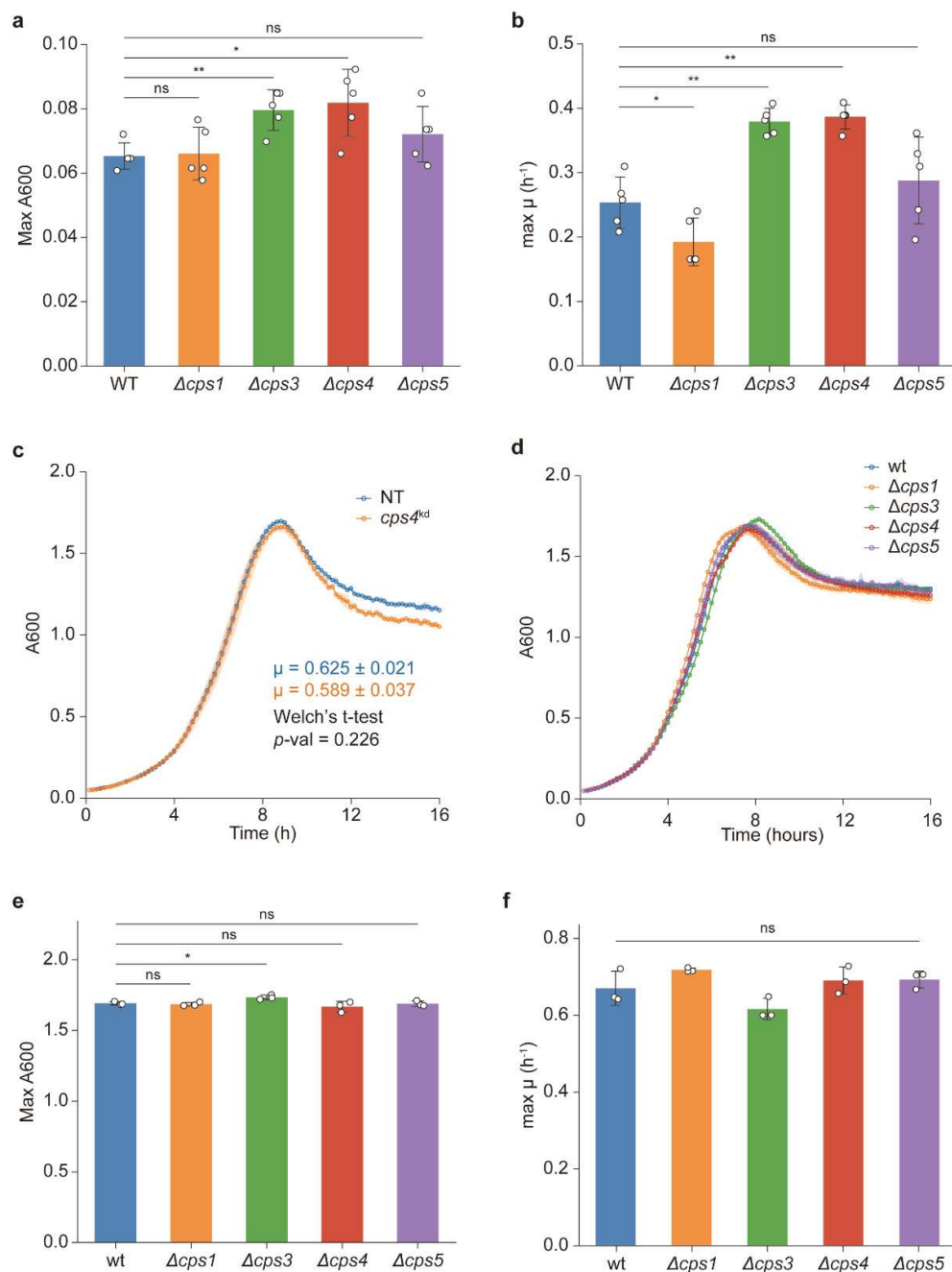


Figure S5. Growth phenotypes of *cps* mutants and *cps4*-repressed strains. . Measured growth parameters of (a) maximum growth rates and (b) specific growth rates between WT and Δcps strains in MM-muc. (c) Growth kinetics of NT control versus $cps4^{kd}$ strain in MM-glc. (d) Growth kinetics of WT versus four *cps* deletion strains. Measured growth parameters of (e) maximum growth rates and (f) specific growth rates between WT and Δcps strains. Significance was determined by Welch's t-test. Asterisk represents statistical significance * p -val < 0.05; ** p -val < 0.01.

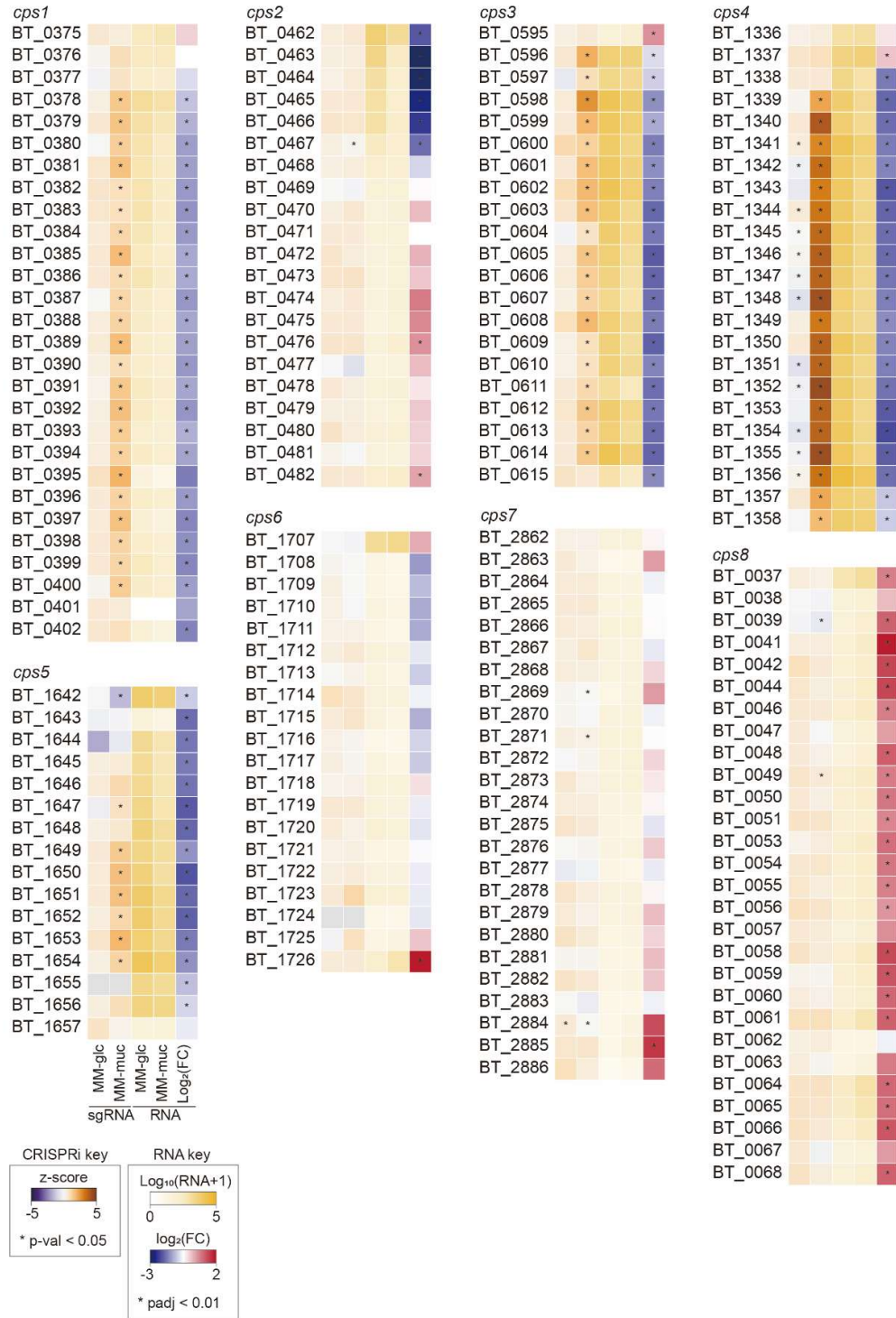


Figure S6. Fitness and transcriptional profiles of the eight capsular *cps* loci in MM-glc and MM-muc. Heatmaps showing gene-level CRISPRi fitness scores and transcript abundance for all annotated genes within the eight capsular polysaccharide (*cps1*–*cps8*) loci

of *B. thetaiotaomicron*. For each locus, the first two columns represent CRISPRi-derived fitness z-scores in MM-glc and MM-muc, respectively, and the next three columns represent RNA expression in MM-glc, MM-muc, and the corresponding $\log_2$ fold change ($\log_2(\text{FC})$) between the two conditions. Asterisks denote statistical significance ($p < 0.05$ for CRISPRi; adjusted $p < 0.01$ for RNA-seq).

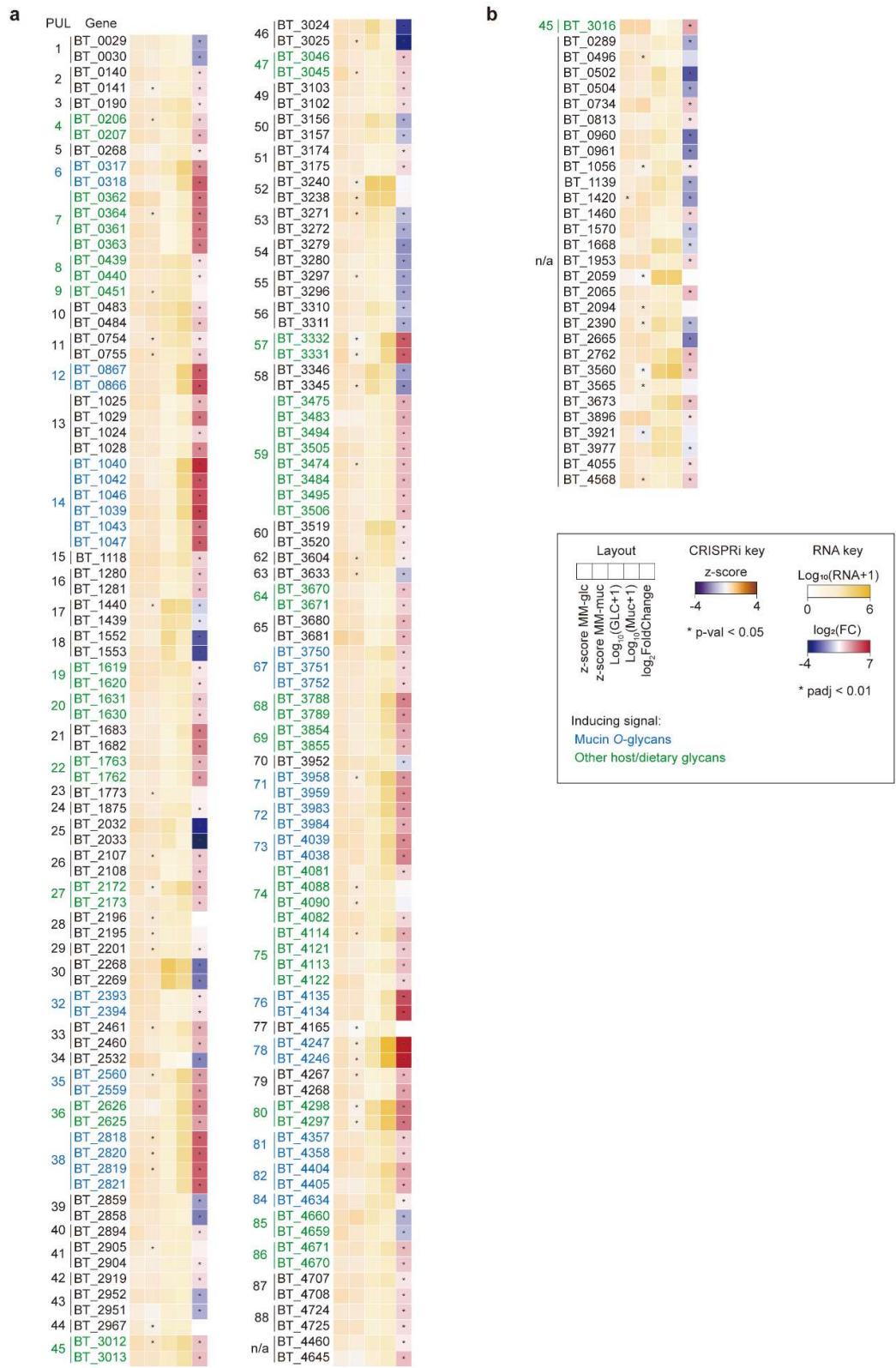


Figure S7. Heatmap displaying genes predicted to encode other membrane-bound, glycan sensing and transport machineries. (a) SusCD and (b) TonB-dependent transport systems. Genes significantly changed in terms of either fitness (z-score with p -val < 0.05; Mann-Whitney U) or expression (DEGs with $padj$ < 0.01) are displayed. Asterisks (*) indicates statistical significance.

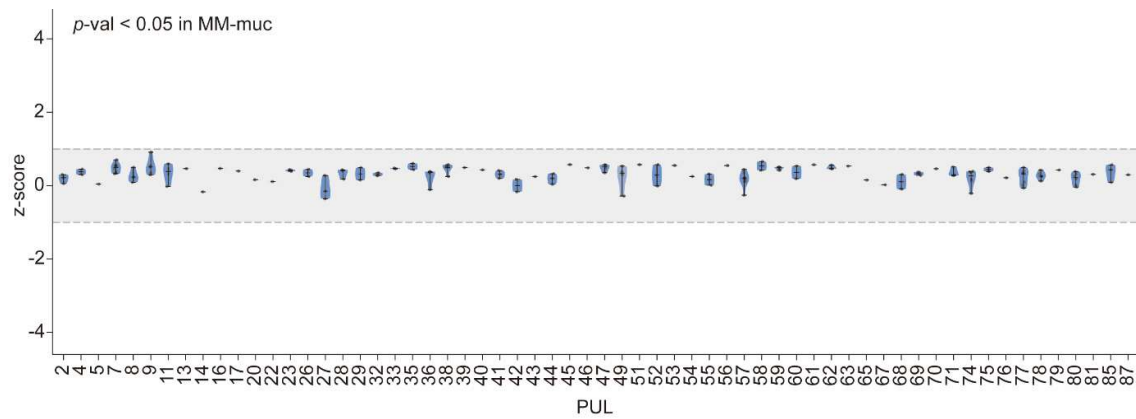


Figure S8. Distribution of fitness z-scores for genes categorized by PULs statistically significant in MM-muc. Violin plots show the spread of z-scores for individual genes within each family. Genes with $|z\text{-score}| > 1.0$ are highlighted.

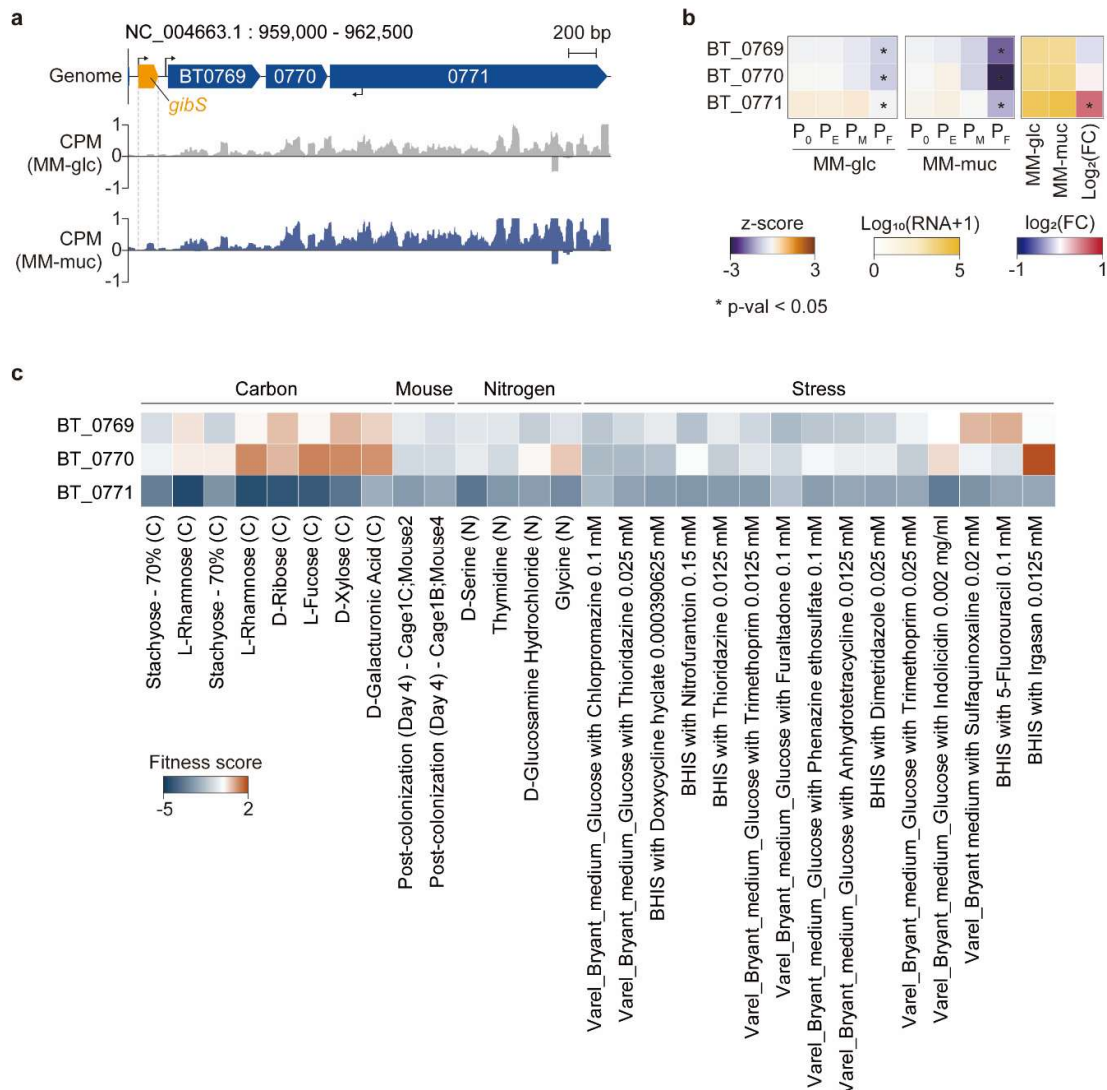


Figure S9. Genomic context, expression, and condition-dependent fitness of the *gibS*–BT_0769–BT_0771 locus. (a), Genomic organization of the *gibS* locus and adjacent genes BT_0769, BT_0770, and BT_0771 in *B. thetaiotaomicron*. RNA-seq coverage profiles, shown as counts per million (CPM), are displayed for cells grown in MM-glc or MM-muc. (b) Heatmap showing CRISPRi-derived fitness z-scores for BT_0769, BT_0770, and BT_0771 across sequential passages in MM-glc and MM-muc, together with corresponding RNA abundance and differential expression between conditions. Asterisks indicate statistically significant values ($p < 0.05$). (c) Heatmap of fitness scores for BT_0769, BT_0770, and BT_0771 across diverse environmental conditions, grouped by carbon source, mouse colonization, nitrogen source, and stress-related perturbations.

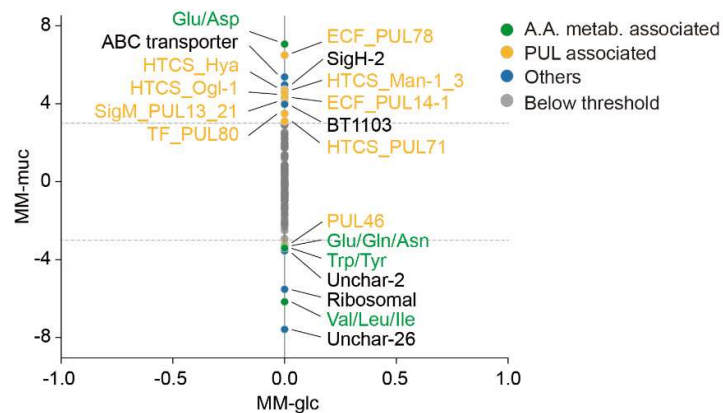


Figure S10. Comparison of iModulon activities between MM-glc and MM-muc. Dotted lines indicate activity thresholds (3 and -3, respectively). iModulons with distinct functional categories are highlighted in different colors. Abbreviations: A.A. metab. – amino acid metabolism; PUL: polysaccharide utilization loci.

regulation of key cellular modules. iModulons with distinct functional categories are highlighted in unique color codes. Those with thick red or blue borders indicate iModulons with relatively higher (iModulon activity > 1) or lower activity (iModulon activity < -1) respectively in MM-muc than in MM-glc. For multi-iModulon boxes with contrasting iModulon activity (containing iModulons with both higher or lower iModulon activity) were highlighted in green.

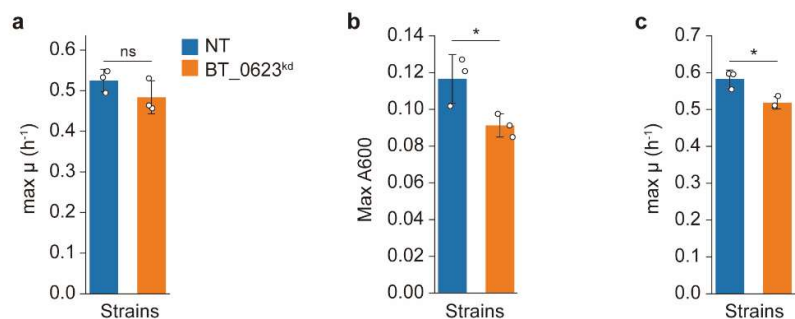


Figure S12. Comparison of growth phenotypes between NT and BT0623-repressed strains. (a) Specific growth rates in MM-glc. **(b)** maximum cell density in MM-muc. **(c)** Specific growth rates in MM-gal.

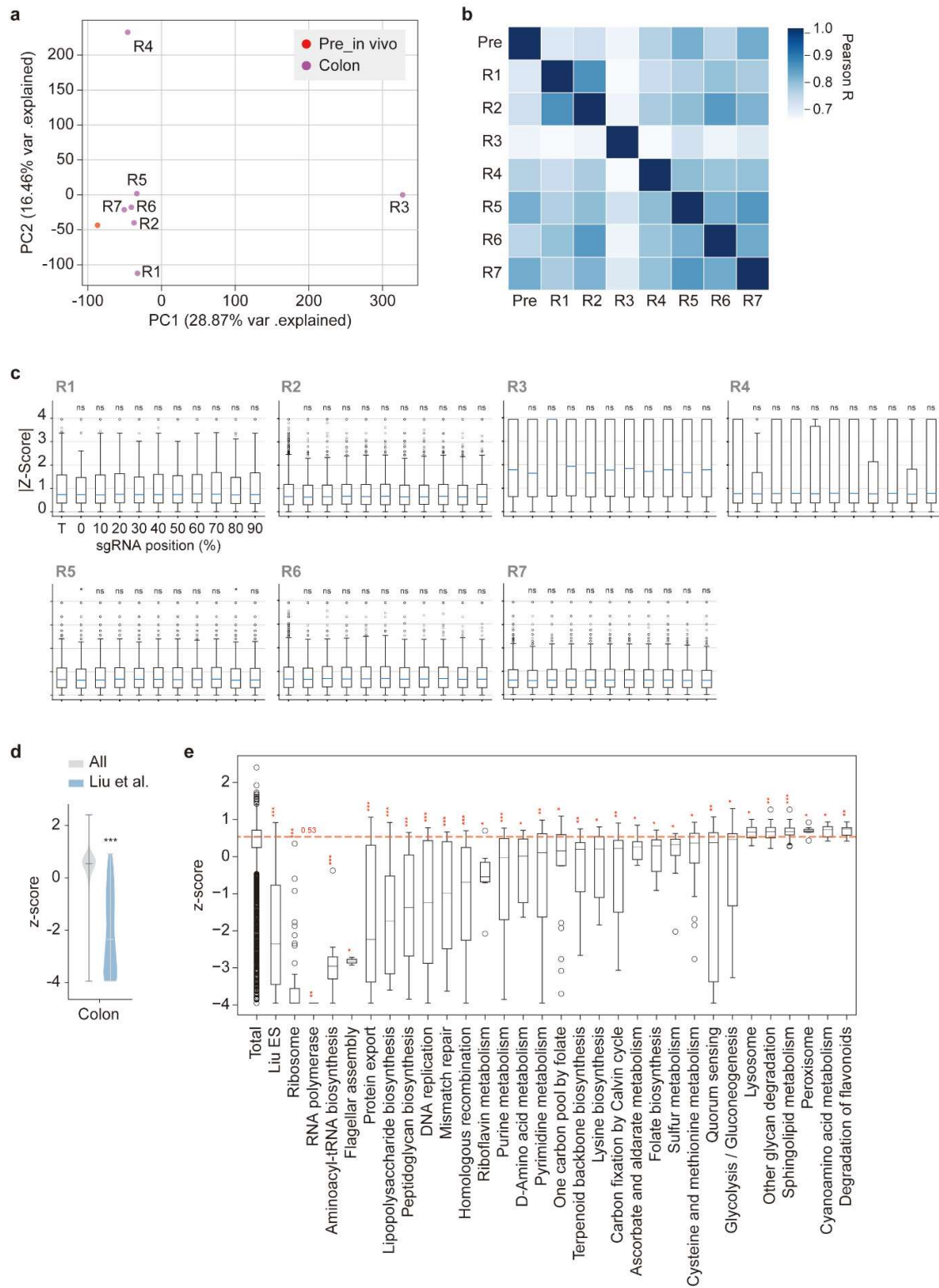


Figure S13. Quality assessment and functional validation of the in vivo CRISPRi screen during murine colonization. (a) PCA of genome-wide sgRNA abundance profiles comparing the input library (Pre_in vivo) with biological replicates recovered from the murine colon (R1–R7). (b) Heatmap showing pairwise Pearson correlation coefficients of genome-wide sgRNA abundance profiles among the input library and colon replicates. (c) Distribution of sgRNA-derived absolute z-scores as a function of relative target position along CDS for each colon replicate. Box plots show the median and interquartile range, with whiskers extending to 1.5× the interquartile range. ns: non-significant. (d) Violin plots comparing the distribution of gene-level fitness z-scores for all genes versus the previously defined essential-gene [14]. (***, $p < 0.001$). (e) Box plots showing the distribution of gene-level z-scores across selected KEGG functional categories in the colon CRISPRi screen.

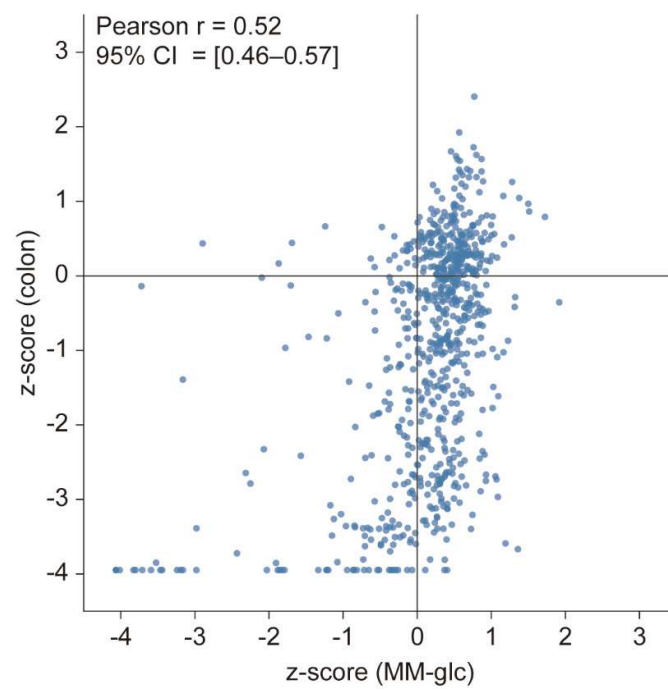


Figure S14. Z-score correlation plot between MM-glc and in vivo (colon).

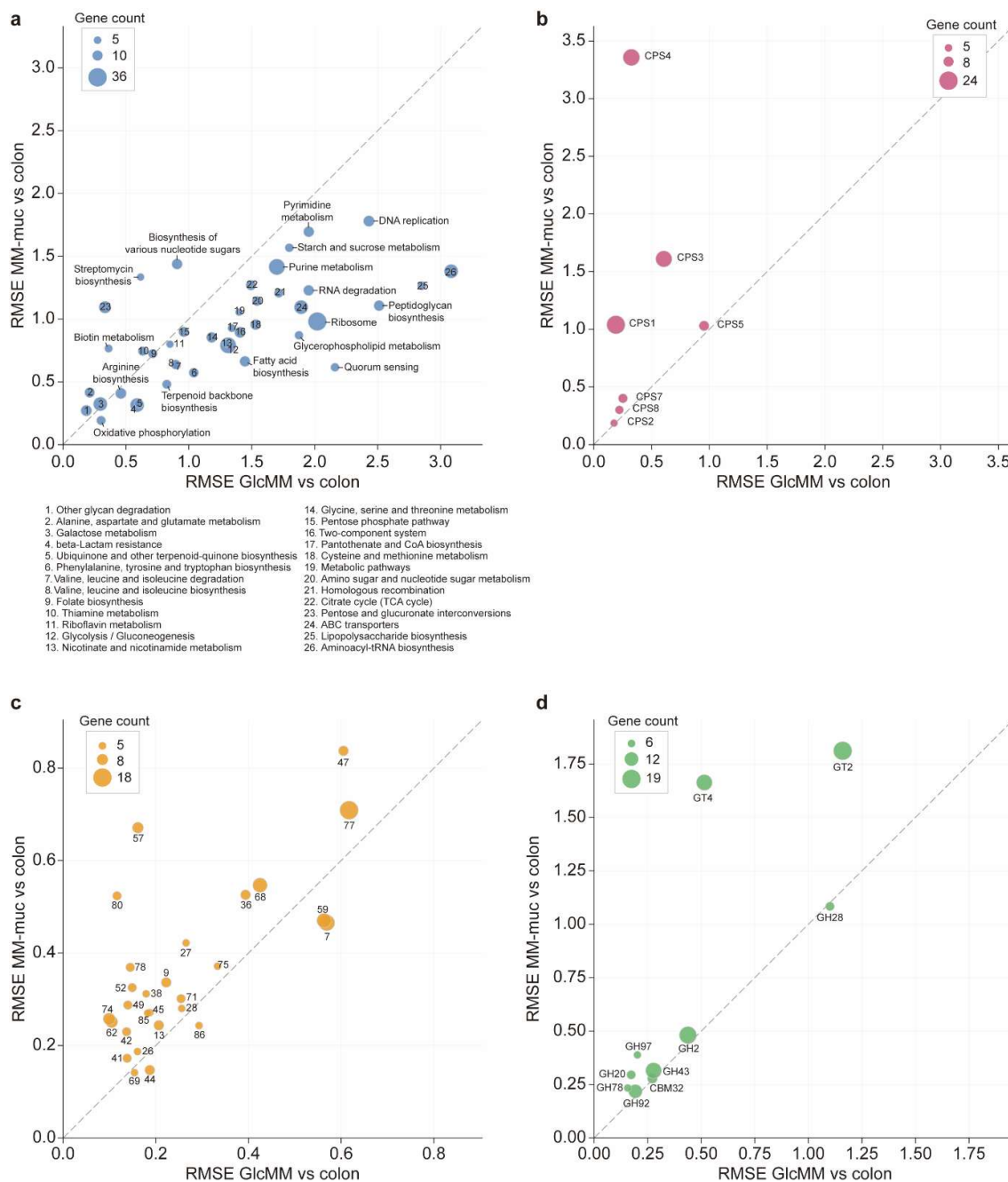


Figure S15. Category-level root mean square error (RMSE) comparison of in vitro fitness profiles against the in vivo colon profile. (a) KEGG categories. (b) CPS loci. (c) PULs. (d) CAZy families. For each functional subcategory, the RMSE between MM-muc and colon z-scores is plotted on the y axis, and the RMSE between GlcMM and colon z-scores is plotted on the x axis. Each point represents one subcategory, and circle size indicates the number of genes assigned to that subcategory. The dashed diagonal denotes equal RMSE between the two *in vitro* conditions.

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